

ETF Portfolio Optimization

Maximizing Sharpe Ratio with Analyst-Supplied Expected Returns

7-ETF Portfolio | SLSQP Optimization | Analyst Forward-Looking μ

Claude Generated

Methodology at a Glance



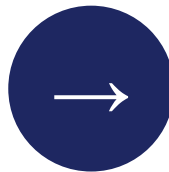
Expected Returns

Analyst-supplied
forward-looking views
(NOT historical means)



Covariance Matrix

Historical daily log returns ×
252 trading days
(2021–present)



Objective

Maximize Sharpe Ratio via
SLSQP solver fully invested,
4%–24% per ETF

Risk-Free Rate: 5.00% | Benchmark: Equal-Weight (1/N)

Asset Universe & Analyst Return Assumptions

Ticker	Asset Class	Analyst μ	Hist. Vol	Implied Sharpe
IWV	US Equity (Russell 3000)	8%	~14%	0.21
IEMG	Emerging Market Equity	8%	~17%	0.18
VEA	Developed Non-US Equity	7.9%	~14%	0.21
JNK	High-Yield Bond	6.6%	~9%	0.18
SPTL	Long-Term Government Bond	5.1%	~15%	0.01
VCIT	Investment-Grade Corp Bond	4.9%	~7%	-0.01
TIP	Inflation-Protected Bond	4.7%	~8%	-0.04

Portfolio Performance Results

Equal-Weight Baseline

6.64%

Expected Return

9.32%

Volatility

0.178

Sharpe Ratio

Max-Sharpe Optimized

7.11%

Expected Return

+0.47%

8.82%

Volatility

-0.50%

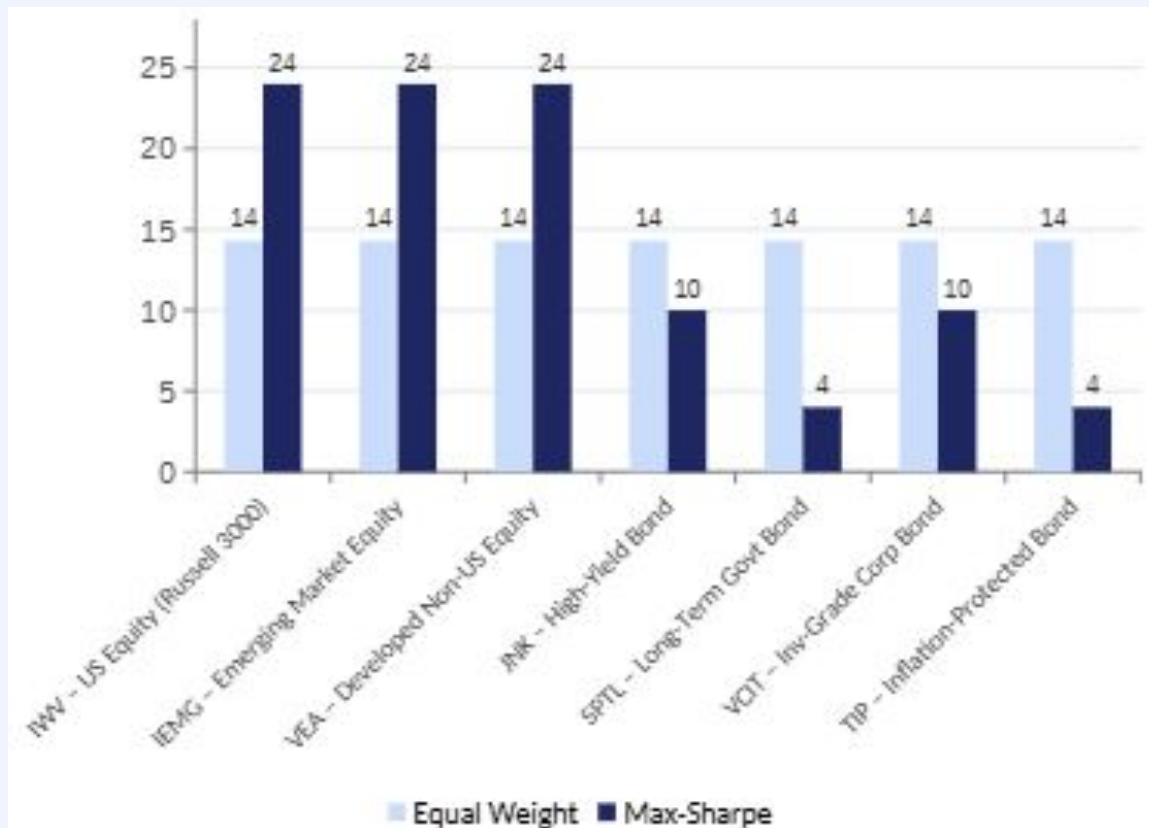
0.238

Sharpe Ratio

+0.060

33.7% Sharpe Improvement vs Equal-Weight Baseline

Optimal Portfolio Weights



MAX ▲ 24%

IWV / IEMG / VEA

MIN ▼ 4%

SPTL / TIP

FREE ○

JNK / VCIT

Efficient Frontier — Key Portfolios



Global Min. Variance

5.18%

Return

7.21%

Vol

0.025

Sharpe



Tangency (Max Sharpe)

7.11%

Return

8.82%

Vol

0.238

Sharpe



Equal Weight (1/N)

6.64%

Return

9.32%

Vol

0.178

Sharpe

Tangency portfolio sits on the Capital Market Line — the optimal risk/return trade-off given the analyst return assumptions.

Optimal Weights

- Maximum weight (24%): IWV, IEMG, VEA
- Minimum weight (4%): SPTL, TIP
- Free weight (10%): JNK, VCIT